ASCII MASTER FLOW — PANEL 1/8: Industrial geography: terminology and location question

```
+---------------- INDUSTRIAL GEOGRAPHY ----------------+
| CENTRAL QUESTION: Why does an activity locate here,  |
| cluster there, persist, disperse or relocate?         |
+----------------------------------------------------------+
                         |
INDUSTRY = broad organised economic activity
MANUFACTURING = transformation of material into products
FACTORY = individual production establishment
INDUSTRIAL REGION = dense interacting firms, labour and infrastructure
                         |
LOCATION FACTOR SYSTEM
raw material + energy + water + labour + capital + market
+ transport + land + technology + policy + agglomeration
                         |
firm choice -> supplier links -> labour settlement -> shared services
                         |
cluster growth or failure depends on linkage quality and external costs

Do not use factory, manufacturing and industrial region as synonyms.
```

ASCII MASTER FLOW — PANEL 2/8: Ranked location factors and industrial orientation

```
FIRST ASK: Which cost, risk or capability dominates this industry?
                              |
       +----------------------+----------------------+
       |                      |                      |
RAW-MATERIAL            MARKET                 FOOTLOOSE / SKILL
weight loss             perishability          high value-to-weight
bulk or fragility       custom demand          knowledge and networks
       |                      |                      |
ore processing          food/bakery             electronics / IT-linked
       +----------------------+----------------------+
                              |
PORT ORIENTATION: imported input + export market + deep logistics
POWER ORIENTATION: energy-intensive process
LABOUR ORIENTATION: skill, wage, productivity and labour ecosystem
STATE ORIENTATION: serviced land, trunk utility and policy support
                              |
RANK FACTORS BY PROCESS STAGE
input -> transformation -> assembly -> distribution -> after-sales

A multi-stage value chain can split across several locations.
```

ASCII MASTER FLOW — PANEL 3/8: Weber's least-cost theory in six analytical steps

```
1. LOCATE raw-material sources and market
                         |
2. CLASSIFY inputs: ubiquitous or localised; pure or weight-losing
                         |
3. COMPARE transport burden through the material index
                         |
4. FIND transport-minimum point within the locational triangle
                         |
5. TEST labour saving against added transport cost
                         |
6. TEST agglomeration saving against congestion and dispersion cost
                         |
                 LEAST TOTAL COST LOCATION

TRANSPORT COST + LABOUR COST + AGGLOMERATION ECONOMY
                         |
MODERN QUALIFICATION
containers + roads + reliable power + data + regulation + value chains
alter cost surfaces, but distance, weight, time and reliability still matter.

Weber explains a baseline mechanism, not every final location.
```

ASCII MASTER FLOW — PANEL 4/8: Major world industrial regions and their geographic mechanisms

REGION	MECHANISM
NE United States	market + Great Lakes + transport + skills
Ruhr / NW Europe	coal legacy + Rhine links + dense markets
Midlands / N Italy	old industry + SMEs + European networks
Japan Pacific Belt	ports + imported inputs + coastal markets
China coastal belt	ports + labour + FDI + supplier depth
South Korea corridor	ports + chaebol networks + technology

COMMON EVOLUTION

resource or port advantage -> firm concentration -> suppliers and labour
-> infrastructure and knowledge -> diversification -> industrial inertia

RESTRUCTURING PRESSURE

old plant + pollution + land cost + global competition
-> upgrading, decline, relocation or service transition

The same label hides different factor combinations and historical paths.

ASCII MASTER FLOW — PANEL 5/8: India's layered industrial map: belts, clusters and corridors

```
INDIA'S LAYERED INDUSTRIAL GEOGRAPHY
Chota Nagpur / Damodar : coal + ore + rail -> steel, power, engineering
Mumbai-Pune            : port + market + capital -> diversified industry
Ahmedabad-Vadodara     : cotton + enterprise -> textile, chemical, engineering
Chennai-Bengaluru      : ports + skill -> auto, electronics, machinery
Coimbatore-Tiruppur    : textile base + SMEs -> knitwear, pumps, machinery
Delhi-NCR              : market + policy + logistics -> diversified cluster
Visakhapatnam/east     : port + mineral hinterland -> metal and petro industry
                         |
CORRIDOR OVERLAY
freight spine / expressway / port -> planned node -> trunk utilities
-> firms and suppliers -> labour settlements -> possible industrial region
                         |
DMIC west | AKIC east | Chennai-Bengaluru | Vizag-Chennai

New corridors overlay inherited belts; they do not erase brownfield depth.
```

ASCII MASTER FLOW — PANEL 6/8: Agglomeration, deglomeration, inertia and regional divergence

```
AGGLOMERATION LOOP
firms cluster -> shared suppliers + labour + services + knowledge
-> lower transaction cost -> more firms and skill -> deeper cluster
                         |
                 CONGESTION THRESHOLD
land cost + traffic + pollution + water stress + labour pressure
                         |
DEGLOMERATION
selected plants or stages move outward to cheaper connected sites
                         |
INDUSTRIAL INERTIA
sunk capital + supplier ties + skill + reputation keep old region active

REGIONAL OUTCOME TEST
SPREAD: local suppliers + training + demand + infrastructure diffusion
BACKWASH: talent/capital drain + enclave links + land displacement
                         |
A large plant creates convergence only when local capability and linkages grow.
```

ASCII MASTER FLOW — PANEL 7/8: State, corridors, PLI and the industrial intervention stack

```
NATIONAL POLICY: DPIIT / ministries define frameworks and incentives
                         |
CORRIDOR GOVERNANCE: NICDIT / SPVs plan nodes and trunk systems
                         |
STATE AGENCIES: land + utilities + approvals + local industrial estates
                         |
NETWORK ACTORS: rail freight + ports + roads + power + digital systems
                         |
FIRM ECOSYSTEM: anchor producer + suppliers + skills + testing + finance
                         |
URBAN SUPPORT: housing + mobility + water + waste + social services
                         |
OUTCOME: production + jobs + local linkages + lower logistics cost

STATUS FIREWALL
outlay -> application -> investment -> production -> incentive -> wider outcome

PLI may reinforce capable clusters unless weaker regions gain real ecosystems.
A transport line becomes a corridor only through functioning nodes and linkages.
```

ASCII MASTER FLOW — PANEL 8/8: Map-answer and Mains spine for industrial location and regions

```
QUESTION: How is India's industrial geography being reorganised?
                              |
1. DEFINE: factory, manufacturing, cluster, region and corridor
                              |
2. MAP: mineral east | western port belt | southern skill arcs | NCR
                              |
3. EXPLAIN INHERITED LOCATION
   raw material + port + market + labour + transport + agglomeration
                              |
4. ADD NEW NETWORK LOGIC
   freight corridor + planned node + supplier ecosystem + incentives
                              |
5. TEST REGIONAL EFFECT
   jobs + local procurement + skill transfer versus enclave/backwash
                              |
6. QUALIFY STATUS
   announcement, construction, operation and outcome need separate evidence
                              |
7. VERDICT
   designed concentration becomes development only when logistics,
   suppliers, skills, settlements, ecology and institutions work together.
```